

What a Survey Can Establish That No Record Can

What no sample size will buy

84-355 Democracy's Data

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Most of this book reads records. A record is made because somebody was required to make it: a ballot cast, a district drawn, a contribution filed, a person counted. The record exists whether or not anyone ever studies it, and the questions it can answer were fixed by whoever was obliged to write it down.

This chapter is about the other kind of source. A survey exists because somebody **asked**, and it records not what a person *is* but what a person *says* – in the categories the question allowed, to an interviewer, on one day.

That difference is not a limitation to be apologised for. It is the reason surveys exist at all, because there are things about a democracy that no record anywhere contains.

01 Enumeration and elicitation

A census tries to count everyone. A survey asks some people questions. The two operations fail in different places, and neither failure can be fixed by doing more of the other thing.

The question	Census	ANES	CES
How many people live on this block?	Yes, by enumeration	No – a sample cannot count a block	No – a sample cannot count a block
What share of adults in Wyoming are registered?	Yes, by enumeration	No – too few respondents per state	Yes, at about +/- 6 points
Does this person trust the federal government?	No – no record contains an attitude	Yes, and since 1958	Yes, but only back to 2006
Has the country's view of abortion changed since 1980?	No – the question is never asked	Yes – this is what it is for	No – the series is too short
Do Black Republicans differ from white Republicans on immigration?	No – neither variable exists	No – the subgroup is too small	Yes – this is what the sample size is for
What is this person's race, as they describe it?	Yes, in the categories the form allows	Yes, in the categories the question allows	Yes, in the categories the question allows

Read the third row. **No administrative record in the United States contains whether a person trusts the government.** There is no register of opinions, there has never been

one, and a free society is unlikely to build one. If you want to know whether trust in government fell after 1964, there is exactly one way to find out, and it is to have asked people at the time. Somebody did, which is why the question is answerable now and would be unanswerable otherwise.

Then read the first row. A survey cannot tell you how many people live on a block, and this is not a matter of trying harder. It is arithmetic, and the arithmetic is unforgiving in a way that surprises people.

02 What no sample size will buy

Here is the most recent American National Election Study, which is a large, expensive, carefully drawn national survey.

Quantity	Value
Respondents in the study	5,521
States and equivalents represented	53
Respondents in the median state	26
Respondents in the smallest state	2
Respondents in the largest state	629
Margin of error, median state, at 50%	± 19.2 points
Margin of error, smallest state, at 50%	± 69.3 points
Margin of error, whole sample, at 50%	± 1.3 points

5,521 respondents give a national estimate accurate to about **± 1.3 points**, which is excellent. Split those same respondents by state and the median state holds **26 people**, giving **± 19.2 points**. That range is so wide that a result of 50% is consistent with a true value anywhere from a third to two-thirds. The smallest state has 2 respondents and a margin of ± 69.3 points, which is not a measurement at all.

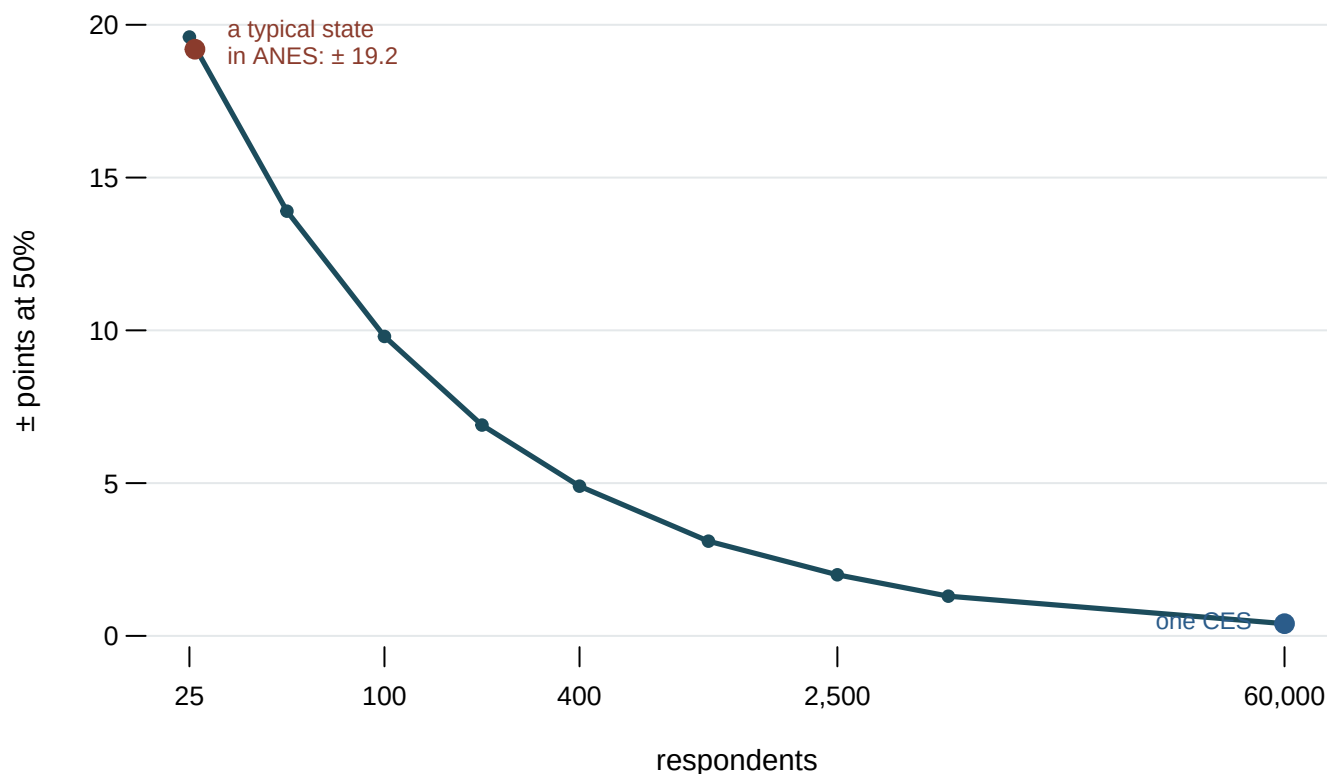


Figure 1. Margin of error at 50% by number of respondents. A margin of error says how far off a sample's answer could be. Move along the curve to read the margin at any sample size. The shape is the whole lesson.

Precision improves with the square root of the sample size, so a national survey cannot be turned into a state survey by analysing it harder. Cutting a sample into fifty pieces costs a factor of about seven. The two marked points are a typical state subsample in ANES and a single CES.

This is the entire reason two national election surveys exist rather than one.

ANES buys depth and time. The same questions since 1948, long interviews, and a design built so that 1972 and 2024 can be set beside each other. It is the only instrument that can answer *has this changed in seventy years*.

CES buys breadth. About sixty thousand respondents in one year. Split the same way, its median state would hold about three hundred people and a margin near ± 6 points. Still not wonderful, but it is the difference between a number you can publish and one you cannot. That is what makes state-level and small-subgroup work possible, and it is why the fifth row of the table above belongs to CES alone.

Neither is better. They are different purchases with the same currency, and a researcher who wants both depth and breadth has to want them in separate projects.

Before you read on, commit to a view. You want to know whether Black Republicans differ from white Republicans on immigration. **Which instrument can answer that, and what would go wrong if you used the other one?**

03 Three decisions between an answer and a number

Nothing in a survey is automatic. Between a person saying something and a percentage appearing in a newspaper there are three decisions, and each one is capable of moving the result more than the thing being measured.

One: the wording is the measurement

Three questions, exactly as the interviewer reads them. Each one is chosen because the wording is not a window onto an opinion -- it is the instrument that produces one.

---- 1. The escape hatch is inside the question -----

We hear a lot of talk these days about liberals and conservatives. Here is a 7-point scale on which the political views that people might hold are arranged from extremely liberal to extremely conservative. Where would you place yourself on this scale, OR HAVEN'T YOU THOUGHT MUCH ABOUT THIS?

The last clause invites a fifth of respondents to decline, and the codebook files that answer as VALID. A survey that omitted the clause would report a country with no such group in it.

---- 2. One column, three questions -----

Generally speaking, do you usually think of yourself as a Republican, a Democrat, an Independent, or what?

...if Republican or Democrat:

Would you call yourself a STRONG [R/D] or a NOT VERY STRONG [R/D]?

...if Independent or other:

Do you think of yourself as CLOSER to the Republican or Democratic party?

The famous seven-point scale is the record of a branch. Nothing in the seven numbers says which branch a respondent came down.

---- 3. The options are the opinion -----

- 1 By law, abortion should never be permitted.
- 2 The law should permit abortion only in case of rape, incest, or when the woman's life is in danger.
- 3 The law should permit abortion for reasons other than rape, incest, or danger to the woman's life, but only after the need for the abortion has been clearly established.
- 4 By law, a woman should always be able to obtain an abortion as a matter of personal choice.

There is no fifth option, and a person whose view is not on the card must pick the nearest one. Whatever share option 3 receives is a fact about the card as much as about the country.

The temptation is to treat a question as a window – as though the opinion were already there and the question merely lets you see it. These three show why that is wrong.

The ideology question carries its own exit, and a fifth of respondents take it. Remove the clause and that group vanishes into the answers, and a country that has one appears to be a country that does not. The party question is three questions wearing one column, so the famous seven-point scale is a record of which branch a person came down – a fact the numbers do not carry. And the abortion card has four options and no fifth, so a person whose view is not printed must choose the nearest thing. **Whatever share the third option receives is a fact about the card as much as about the country.**

Each of these has a chapter of its own: ideology, party identification, and abortion opinion. This chapter's point is the one they share – the instrument is not neutral, and reading the question text is not optional background. It is reading the variable definition.

Two: who answers is not who lives here

People who agree to take surveys differ from people who do not, in ways that correlate with politics. The standard repair is weighting, which means counting some answers more than others: people who resemble under-represented groups count for more than one person.

The size of that repair is worth seeing.

Variable	Category	Interviews	Raw	Weighted	Adjustment
votereg	Yes	54,913	91.5%	72.5%	-19.0
votereg	No	4,424	7.4%	23.2%	+15.8
pid7	Strong Democrat	16,576	27.6%	21.9%	-5.7
newsint	Most of the time	29,960	51.2%	46.4%	-4.8
educ	No HS	2,133	3.6%	7.5%	+3.9
ideo5	Not sure	3,603	6.0%	9.2%	+3.2
votereg	Don't know	663	1.1%	4.3%	+3.2
educ	Some college	13,961	23.3%	20.4%	-2.9
ideo5	Liberal	11,187	18.6%	15.8%	-2.8
gender4	Man	27,454	45.8%	48.4%	+2.6

In the CES 2024 benchmarks, **91.5% of people interviewed said they were registered to vote, and the published estimate is 72.5%** – an adjustment of 19.0 points on a single item.

That is much larger than most of the differences anyone studies with these data. It is not evidence of a problem: it is the correction working, and a survey that reported the raw 91.5% would be badly wrong about the country. But it means the weights are doing a great deal of the work before any result appears, and a reader who does not know they exist is reading a number that is substantially a model.

Three: how many answered decides what may be asked

The third decision is the one in Figure 1, and it is the least negotiable. Every subgroup you name is a smaller sample than the one you started with: a state, a religion, a race within a party. The range grows as you slice. Most published claims about small groups in national surveys are claims the sample cannot support, and the arithmetic to check is a square root.

A second prediction. A pollster reports that support in one state moved six points since last month, from a survey of 500 people in that state. **Should you believe the movement is real?** Use Figure 1 rather than intuition.

04 What this chapter cannot tell you

Margins of error here are the simple ones — 1.96 times the square root of $p(1-p)/n$, at $p = 0.5$, which is the widest case. Real surveys have design effects from clustering and weighting that typically make the true interval larger, sometimes considerably. Every figure in this chapter is therefore **optimistic**, and the honest reading is that things are at least this hard.

The state counts come from where respondents were interviewed in one recent ANES study. Sample designs change between studies, and a different year would give somewhat different per-state counts — but not a different conclusion, because the arithmetic that produces ± 19 points from 26 people does not vary.

The weighting figures describe CES 2024 and are read from benchmarks the ces-class chapter derived. They are not a general claim about how large weighting adjustments usually are. Different surveys, different modes and different years differ, and the registration item is among the largest adjustments in this particular file rather than a typical one.

Finally, this chapter treats “what a person says” as the thing being measured and stops there. Does an answer match a stable underlying attitude, or is it partly invented on the spot in response to the question? That is a real and unsettled argument, and it is the subject of the ANES chapter rather than this one.

05 Sources

- American National Election Studies. *ANES Time Series Cumulative Data File (1948–2024)*, version of 5 February 2026. Ann Arbor, MI and Palo Alto, CA: University of Michigan and Stanford University. <https://electionstudies.org/data-center/anes-time-series-cumulative-data-file/>
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- Ansolabehere, Stephen, Brian Schaffner and Jesse Pope. *Cooperative Election Study, 2024: Common Content*. Harvard Dataverse, doi:10.7910/DVN/X11EP6. Weighting figures via this book's ces-class chapter.

- Groves, Robert M., Floyd J. Fowler, Mick P. Couper, James M. Lepkowski, Eleanor Singer and Roger Tourangeau (2009). *Survey Methodology*, 2nd ed. John Wiley & Sons.
- Zaller, John and Stanley Feldman (1992). "A Simple Theory of the Survey Response." *American Journal of Political Science* 36(3): 579-616.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`capacity.csv`, `instruments.csv`, `moe.csv`, `weighting.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `instruments.csv` puts the same questions to a census and to two surveys. Find a question no survey can answer, and one no census can. What is each instrument actually for?
- `moe.csv` pairs a number of respondents with what that number can support. Find the row where the margin stops being publishable. How many people does a claim actually need?
- `weighting.csv` has `pct_unweighted`, `pct_weighted` and the adjustment between them. Sort by `adjustment`. Which categories does a survey have to correct hardest, and why those?
- `capacity.csv` gives the size of one study. Work out the smallest subgroup you could report from it and still be saying something.